

Name: _____

TRANSFORMATIONS, (ROTATION)

Date: _____

LO: I can rotate shapes about a given centre and describe rotations.

Rotation

A **rotation** turns a shape about a fixed point called the **centre of rotation**. You need to know the centre, the **angle** and the **direction** (clockwise or anticlockwise).

Key rotations: 90° CW: $(x,y) \rightarrow (y,-x)$. 90° ACW: $(x,y) \rightarrow (-y,x)$. 180°: $(x,y) \rightarrow (-x,-y)$. All about the origin.

Quick examples

●	90° CW about (0,0): $(3,1) \rightarrow (1,-3)$	$(x,y) \rightarrow (y,-x)$
●	180° about (0,0): $(3,1) \rightarrow (-3,-1)$	negate both
●	90° ACW about (0,0): $(3,1) \rightarrow (-1,3)$	$(x,y) \rightarrow (-y,x)$
●	90° CW: $(3,1) \rightarrow (-3,-1)$	that is 180°
●	Rotation changes the size of the shape	size stays the same

Key idea

To describe a rotation: state the centre, angle, and direction.
Size and shape are preserved.

90° CW: $(x,y) \rightarrow (y,-x)$ about origin

Every point stays the same distance from the centre

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – 90 CW about origin

Rotate $(2, 5)$ by 90° clockwise about $(0,0)$.

Rule: $(x, y) \rightarrow (y, -x)$

$(2, 5) \rightarrow (5, -2)$

Image is $(5, -2)$

$(5, -2)$

Apply the 90° CW rule: swap and negate the new y.

Example 2 – 180 about origin

Rotate $(-3, 4)$ by 180° about $(0,0)$.

Rule: $(x, y) \rightarrow (-x, -y)$

$(-3, 4) \rightarrow (3, -4)$

Image is $(3, -4)$

$(3, -4)$

180° rotation negates both coordinates.

Example 3 – 90 ACW about origin

Rotate $(4, -1)$ by 90° anticlockwise about $(0,0)$.

Rule: $(x, y) \rightarrow (-y, x)$

$(4, -1) \rightarrow (1, 4)$

Image is $(1, 4)$

$(1, 4)$

90° ACW: negate y then swap.

Example 4 – about another centre

Rotate $(5, 3)$ by 180° about $(2, 1)$.

Translate to origin: $(5-2, 3-1) = (3, 2)$

Apply 180°: $(-3, -2)$

Translate back: $(-3+2, -2+1) = (-1, -1)$

$(-1, -1)$

Subtract centre, rotate, then add centre back.

YOUR TURN – PRACTICE (deliberate practice)

1. 90° CW about origin

Rotate each point 90° clockwise about (0,0).

- a) (1, 3)
- b) (4, 2)
- c) (-2, 5)
- d) (0, -3)

2. 180° about origin

Rotate each point 180° about (0,0).

- a) (3, 1)
- b) (-2, 4)
- c) (5, -3)
- d) (-1, -6)

3. 90° ACW about origin

Rotate each point 90° anticlockwise about (0,0).

- a) (2, 4)
- b) (5, 1)
- c) (-3, 2)
- d) (6, -2)

4. About other centres

Rotate 180° about the given centre.

- a) (4, 3) about (2, 2)
- b) (5, 1) about (3, 3)
- c) (1, 6) about (3, 4)
- d) (7, 2) about (4, 3)

5. Describing rotations

Describe the single rotation that maps A to B (about origin).

a) (2, 3) (3, -2)

b) (4, 1) (-4, -1)

c) (1, 5) (-5, 1)

d) (-3, 2) (2, 3)

e) (6, -2) (-6, 2)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) A rotation preserves the size and shape of a figure ☐

Correct answer: _____

Reason: _____

b) 90° CW about origin: (x,y) (-x, -y) ☐

Correct answer: _____

Reason: _____

c) 180° rotation about origin: (x,y) (-x, -y) ☐

Correct answer: _____

Reason: _____

d) You don't need to state direction for a 180° rotation ☐

Correct answer: _____

Reason: _____

e) 90° CW is the same as 270° ACW ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Triangle PQR has vertices **P(1,1)**, **Q(4,1)**, **R(4,3)**. Rotate it 90° anticlockwise about **(1,1)**. Find the new vertices.

Expression: _____

Simplified: _____

b) A point (a, b) is rotated 90° clockwise about **(2, 3)**. Its image is **(5, 1)**. Find the original point (a, b).

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

